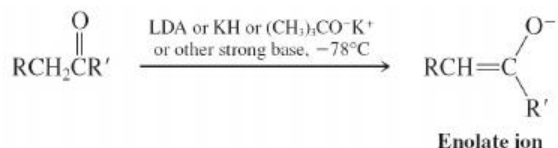


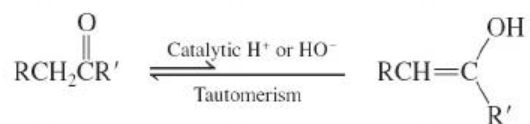
New Reactions

Synthesis and Reactions of Enolates and Enols

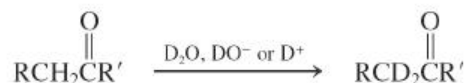
1. Enolate Ions ([Section 18-1](#))



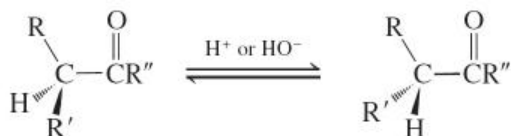
2. Keto–Enol Equilibria ([Section 18-2](#))



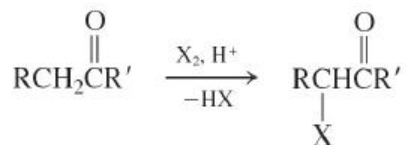
3. Hydrogen–Deuterium Exchange ([Section 18-2](#))



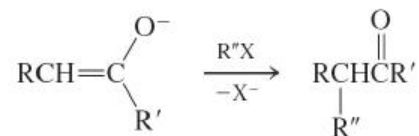
4. Stereoisomerization ([Section 18-2](#))



5. Halogenation ([Section 18-3](#))

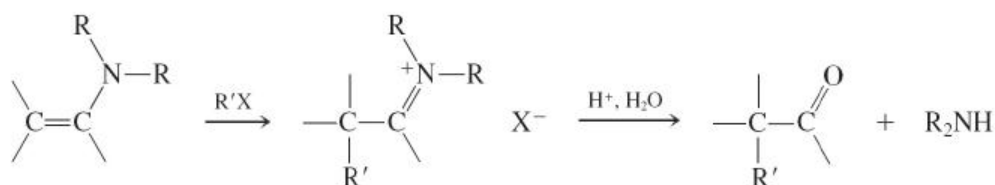


6. Enolate Alkylation ([Section 18-4](#))



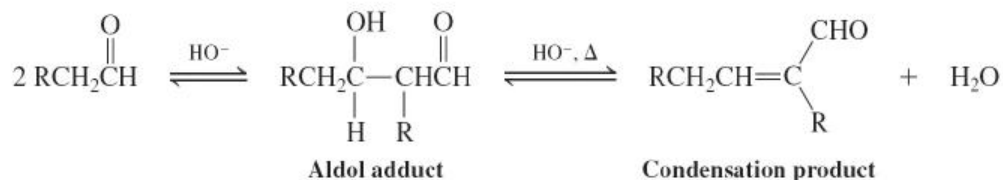
$\text{S}_{\text{N}}2$ reaction: $\text{R}''\text{X}$ must be methyl or primary halide

7. Enamine Alkylation ([Section 18-4](#))

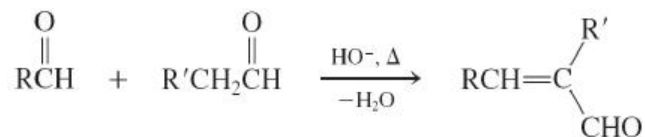


$\text{S}_{\text{N}}2$ reaction: $\text{R}'\text{X}$ must be methyl, primary, or secondary halide

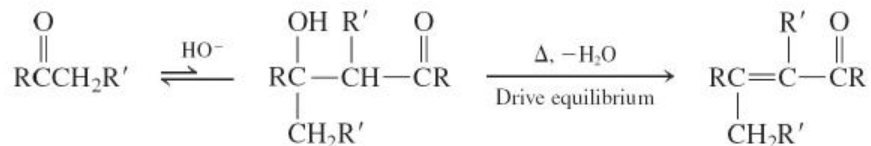
8. Aldol Condensations ([Sections 18-5](#) through 18-7)



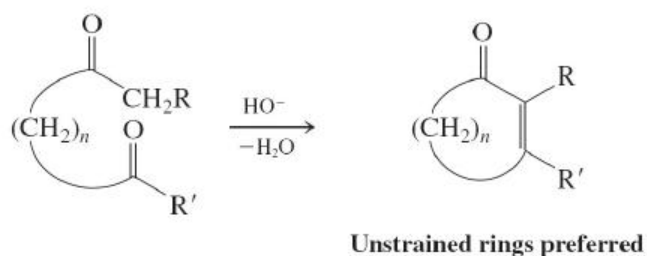
Crossed aldol condensation (one aldehyde not enolizable)



Ketones

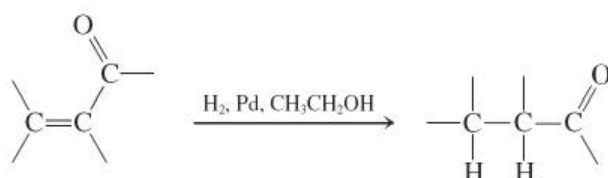


Intramolecular aldol condensation

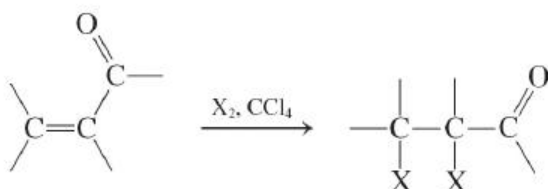


Reactions of α,β -Unsaturated Aldehydes and Ketones

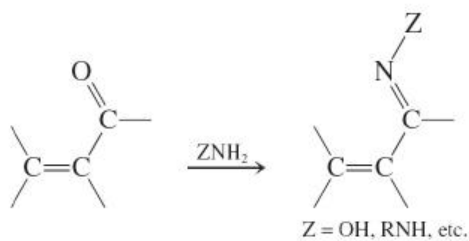
9. Hydrogenation ([Section 18-8](#))



10. Addition of Halogen ([Section 18-8](#))

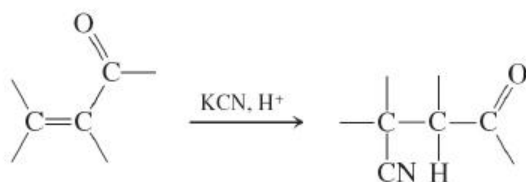


11. Condensations with Amine Derivatives ([Section 18-8](#))

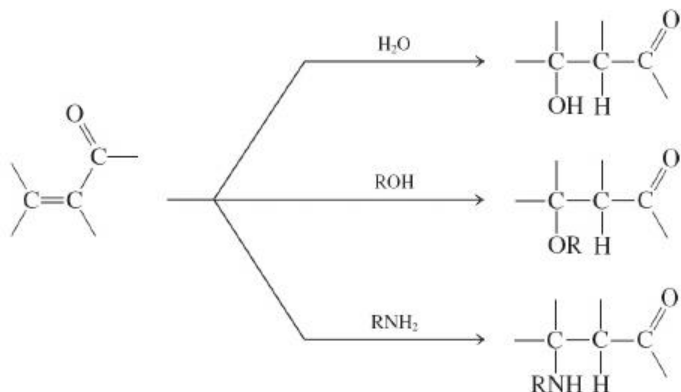


Conjugate Additions to α,β -Unsaturated Aldehydes and Ketones

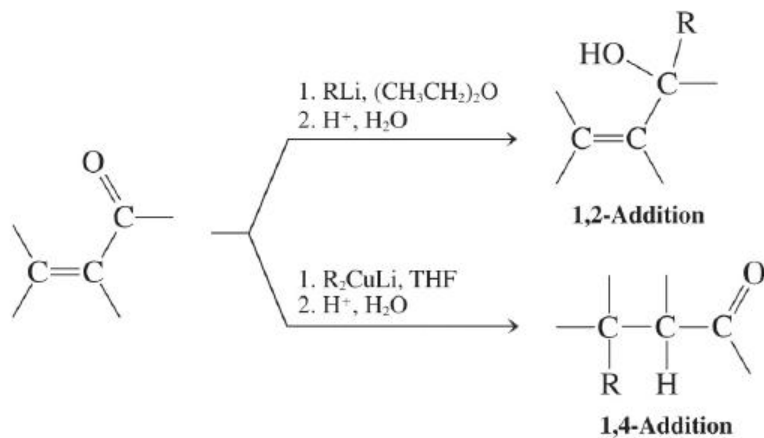
12. Hydrogen Cyanide Addition ([Section 18-9](#))



13. Water, Alcohols, Amines ([Section 18-9](#))

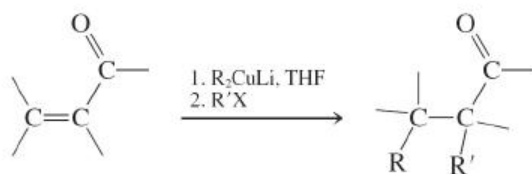


14. Organometallic Reagents ([Section 18-10](#))

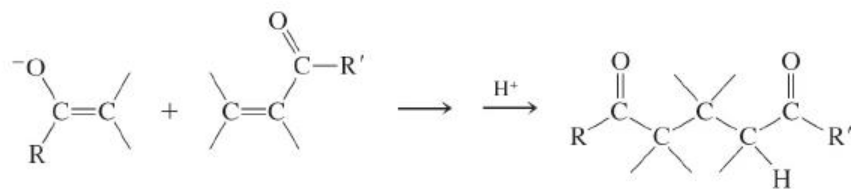


Additions of RMgX may be 1,2 or 1,4, depending on reagent and substrate structure.

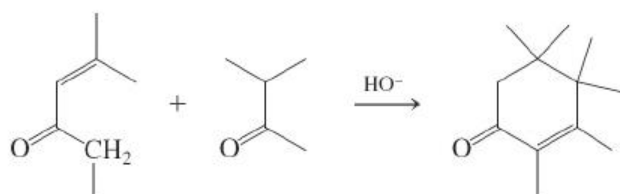
Cuprate additions followed by enolate alkylations



15. Michael Addition ([Section 18-11](#))



16. Robinson Annulation ([Section 18-11](#))



Important Concepts

1. Hydrogens next to the carbonyl group (α -**hydrogens**) are acidic because of the electron-withdrawing nature of the functional group and because the resulting **enolate ion** is resonance stabilized. Unsymmetrical ketones generate the less substituted **kinetic enolate** with LDA at low temperatures or the more substituted **thermodynamic enolate** with a weaker base at higher temperatures under equilibrating conditions.
2. Electrophilic attack on enolates can occur at both carbon and oxygen. Haloalkanes usually prefer α -alkylation. Protonation of the oxygen leads to **enols**.
3. **Enamines** are neutral analogs of enolates. Resonance donation of the nitrogen lone pair imparts nucleophilic character on the remote double bond carbon, which can be alkylated to give iminium cations that hydrolyze to aldehydes and ketones on aqueous work-up.
4. Aldehydes and ketones are in equilibrium with their tautomeric enol forms; the **enol-keto conversion** is catalyzed by acid or base. This equilibrium allows for facile α -deuteration and stereochemical equilibration.
5. **α -Halogenation** of carbonyl compounds may be acid or base catalyzed. With acid, the enol is halogenated by attack at the double bond; subsequent renewed enolization is slowed down by the halogen substituent. With base, the enolate is attacked at carbon, and subsequent enolate formation is accelerated by the halogens introduced.
6. Enolates are nucleophilic and reversibly attack the carbonyl carbon of an aldehyde or a ketone in the **aldol condensation**. They also attack the β -carbon of an α,β -unsaturated carbonyl compound in the **Michael addition**.
7. α,β -Unsaturated aldehydes and ketones show the normal chemistry of each individual double bond, but the conjugated system may react as a whole, as revealed by the ability of these systems to undergo acid- and base-mediated **1,4-additions**. Cuprates add in 1,4-manner, whereas alkyllithiums normally attack the carbonyl function.